Frontiers of Science

Physics and Reality Week 1 Activity

The Relativistic Gamma Factor

In lecture this week you learned about Einstein's famous Special Theory of Relativity, which reveals the intimate relationship between space and time. Professor Greene explained that observers in relative motion would not agree on measurements of distance (space) and duration (time), and showed that there is a simple mathematical relation between the measurements obtained by different observers. **The term relating two different time or distance measurements is referred to as the "relativistic gamma factor."**

$$\gamma \equiv \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$$

where v is the speed of the relative motion and c is the speed of light (3×10^8 m/s).

In this activity, you will practice calculating γ and plot your results on a log plot. Your efforts will reveal some key insights about this "relativistic gamma factor".

Get together in pairs and answer the following questions with your partner.

1. What is the smallest value γ can have? What is the maximum value γ can have? What happens to γ when $v = c$? Explain all answers.

2. Consider a range of speeds, expressed as a fraction of the speed of light. **Calculate the value of γ for the speeds 0.1*c*, 0.5*c*, 0.9*c*, 0.995*c*, 0.99999*c*, and 0.9999995*c*.** (Hint: Keep things in terms of *c* to make the arithmetic easier.)

3. On the graph provided below, plot the values of γ that you obtained in Question 2.

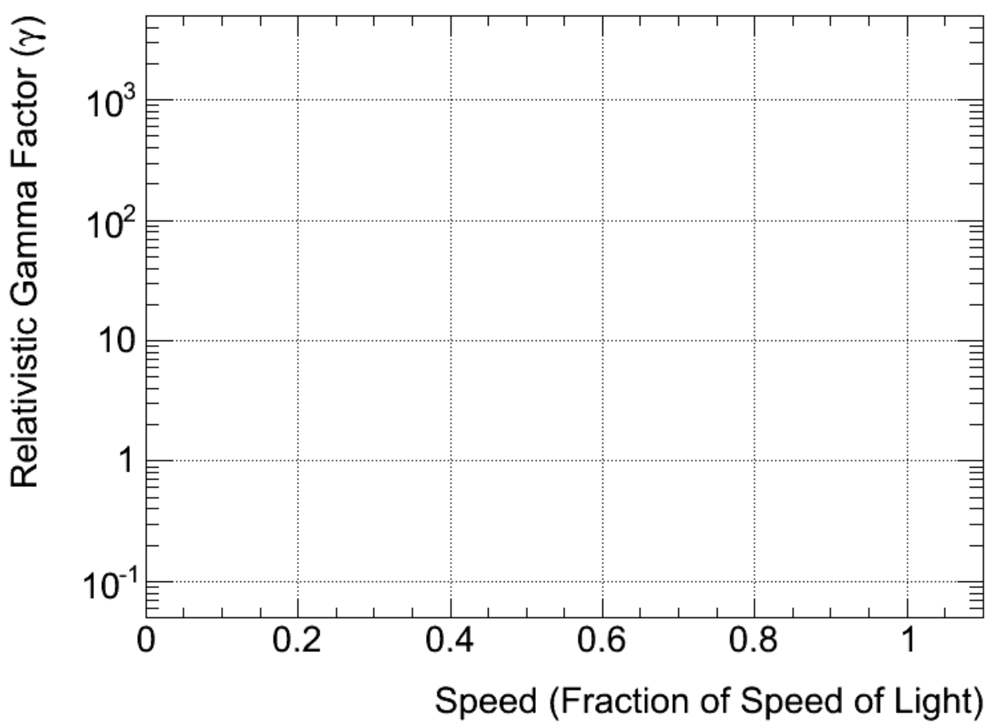

Figure 1. *A semi-log plot of the gamma factor against speed. The scale is logarithmic on the vertical axis, and linear on the horizontal axis.*

4. Does the behavior of the gamma factor, as revealed by the graph in Figure 1, agree with your predictions about how gamma would behave from Question 1? Explain.

5. Using Figure 1, estimate gamma at the following speeds: (i) the speed of a typical commercial transatlantic flight: 500 mph; (ii) the escape velocity from the surface of the Earth: 10,000 m/s. What can you say about the significance of Special Relativity at everyday speeds?

6. Consider t_s, the time elapsed on a clock that is stationary relative to you. **Given t_s, write down an expression for t_m, the time elapsed on a clock that is moving relative to you. Repeat for L_s and L_m, the lengths of an object that is stationary and moving, respectively, relative to you.**